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First Semester, Academic Year, 2026 – 2027



ITEP 414 – System Administration and Maintenance
Week 3 Portfolio Project

Enterprise Server Deployment and Operating System Installation

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1. Project Overview

This project documents the deployment and configuration of an enterprise server operating system in a virtualized environment. Acting in the role of a Junior System Administrator for ABC Startup Solutions, this project covers virtual machine preparation, Ubuntu Server LTS installation, initial server configuration, SSH setup, server verification, BIOS and UEFI firmware concepts, the Linux boot process, a Windows Server Evaluation installation, and a comparison of three enterprise-grade operating systems.

2. Objectives

- Explain the purpose of operating systems in enterprise environments.
- Differentiate BIOS and UEFI firmware.
- Explain the stages of the operating system boot process.
- Compare Windows Server, Ubuntu Server, and Rocky Linux.
- Install Ubuntu Server in a virtual machine.
- Configure the hostname and network.
- Install and verify OpenSSH.
- Verify server connectivity and system updates.
- Document the installation process professionally.



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3. Software Requirements

- Oracle VirtualBox or VMware Workstation
- Ubuntu Server LTS ISO (latest stable version)
- Windows Server Evaluation ISO

4. Virtual Machine Specifications

Component	Specification
VM Name	Ubuntu-Server-Week03
RAM	4 GB
Processor	2 virtual processors
Storage	40 GB
Disk Type	VDI or VMDK
Network	NAT
Operating System	Ubuntu Server LTS

5. Ubuntu Server Installation Procedure

Step 1: Create the Virtual Machine

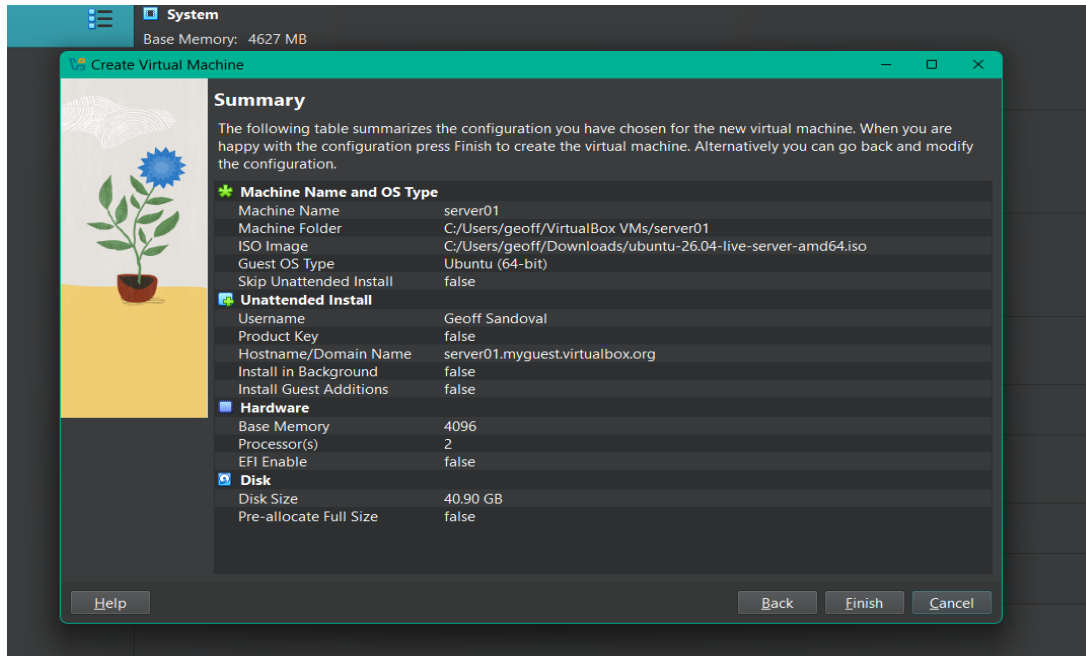
Create a new virtual machine named Ubuntu-Server-Week03, configured with 4 GB RAM, 2 processors,



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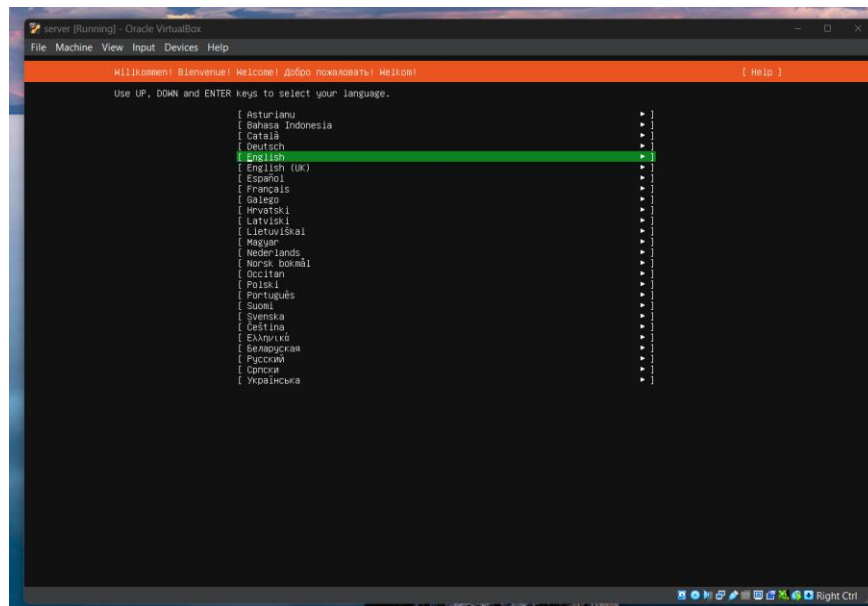


and a 40 GB virtual disk.



Step 2: Select Language

Select English as the installation language.

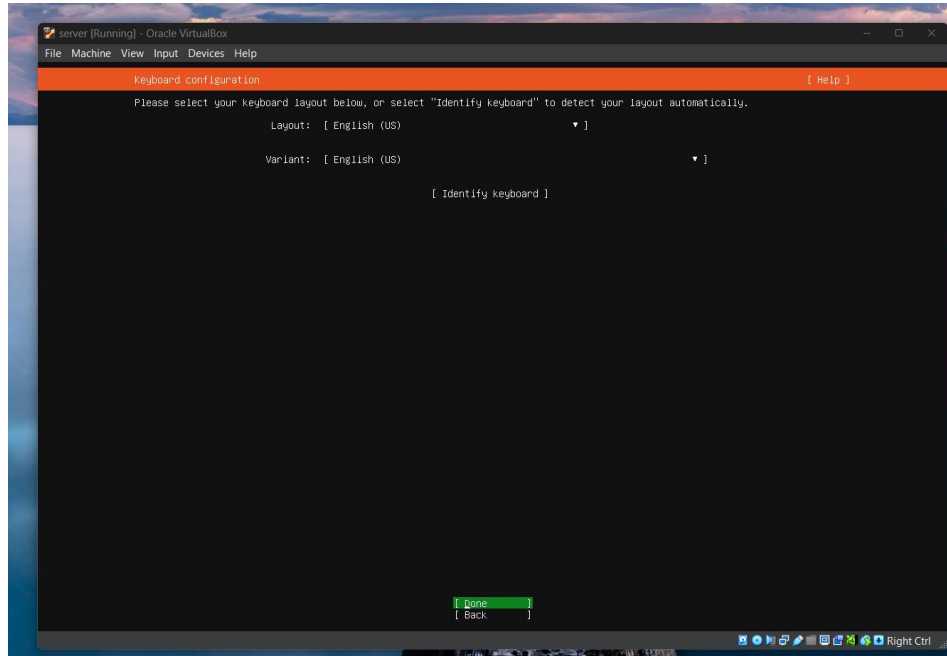




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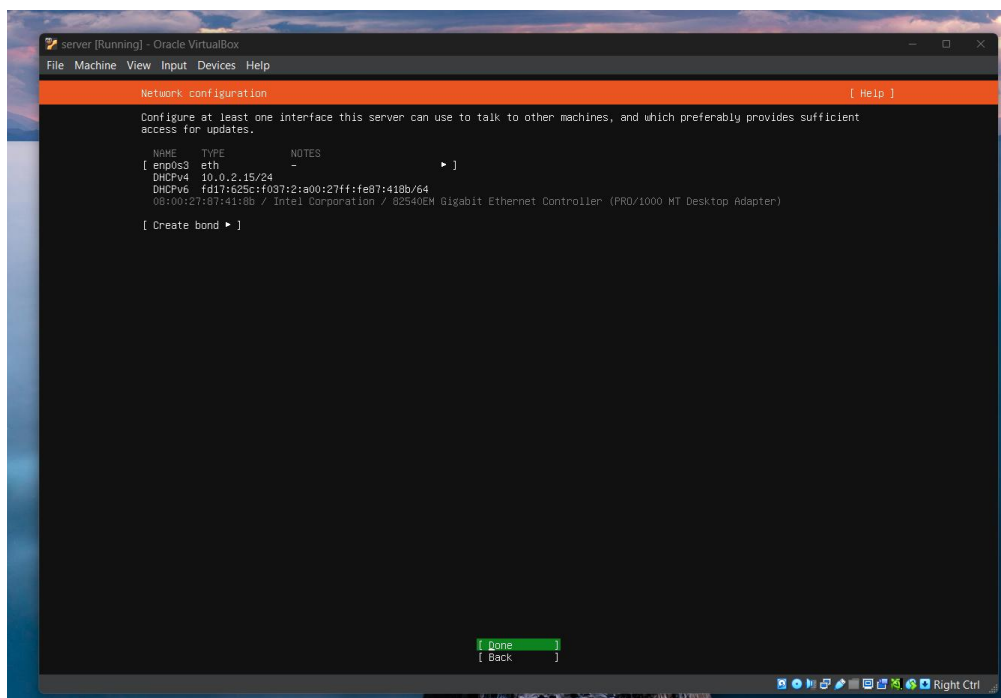


Select the appropriate keyboard layout (e.g., English (US)).



Step 4: Configure Network

Accept DHCP during installation. The address assigned to the server is recorded in Section 6 after installation completes.



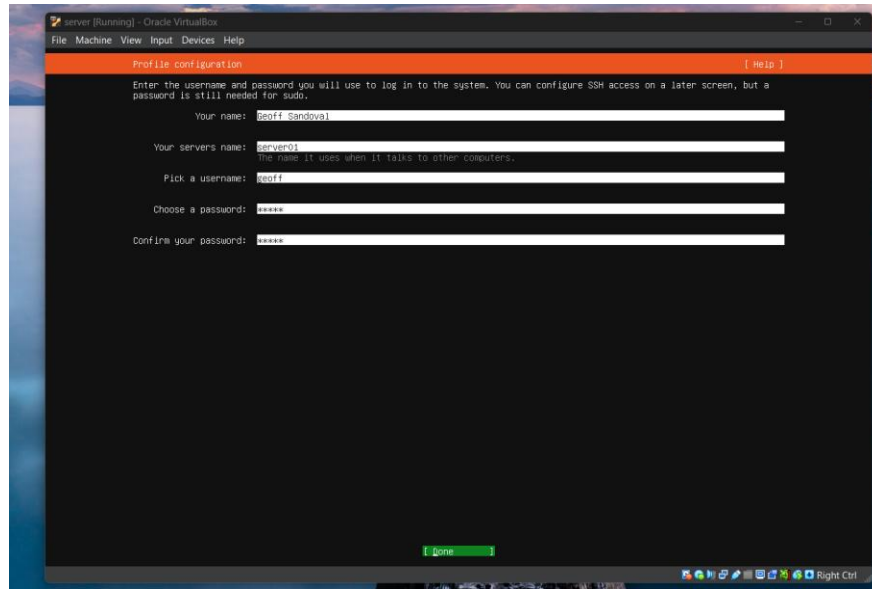


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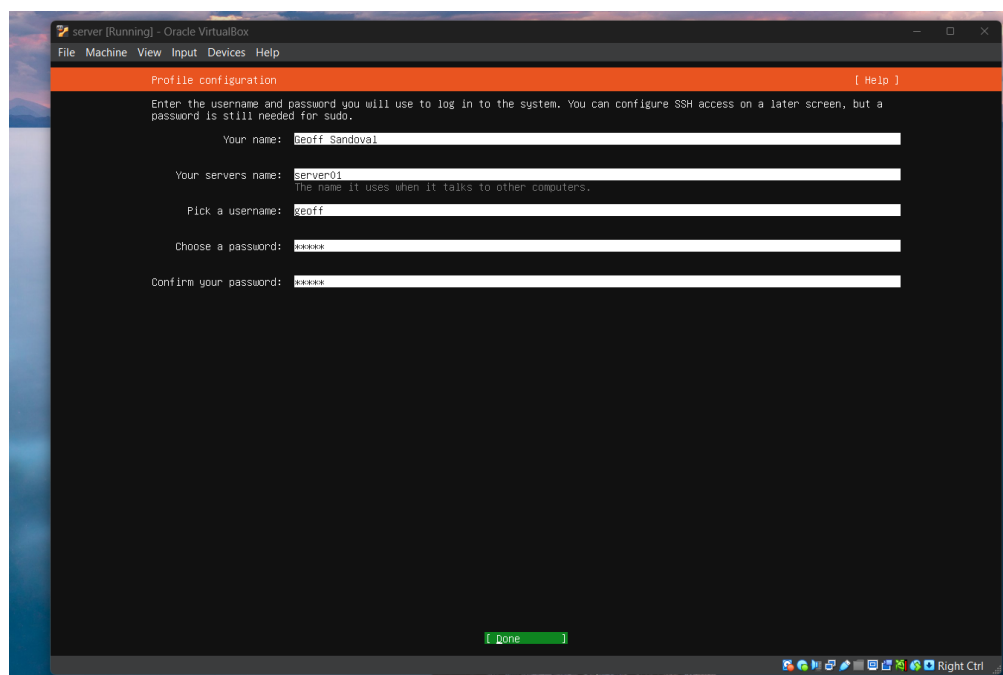
Step 5: Configure Hostname

Set the hostname to server01.



Step 6: Create User Account

Create the required administrative user account and configure a strong password. Passwords are never included in screenshots or documentation.



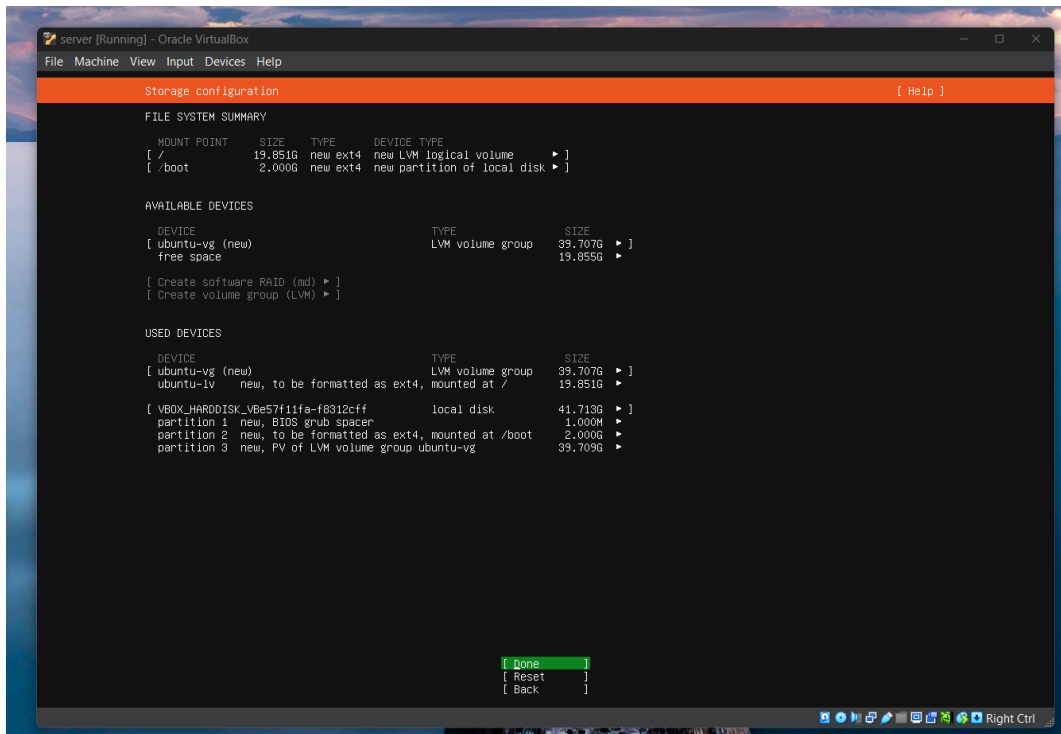


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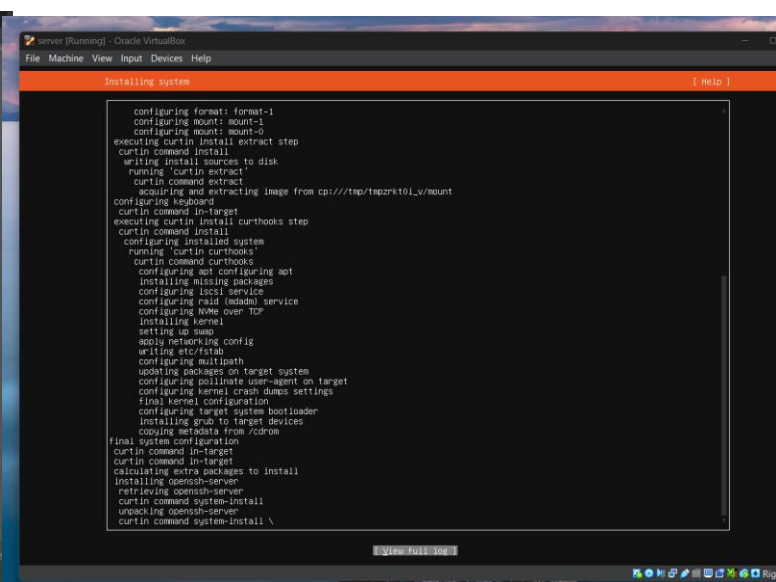
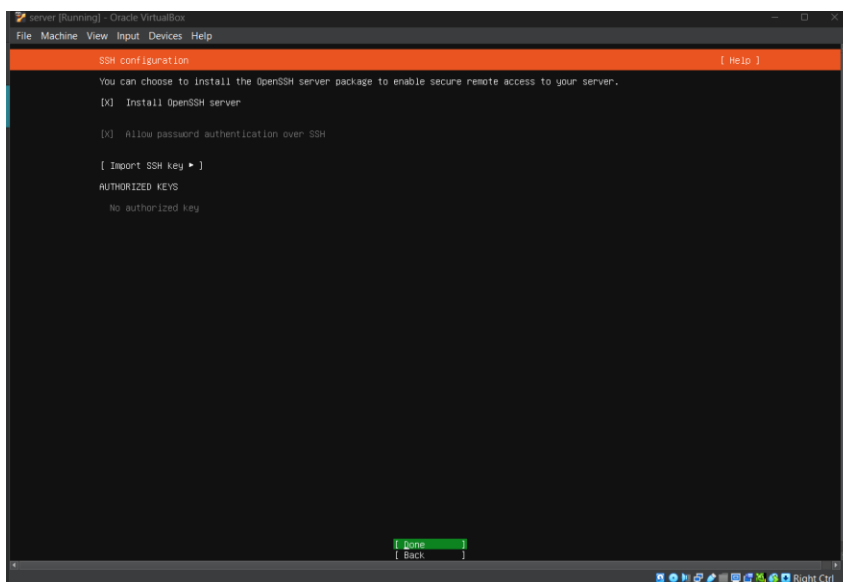
Step 7: Configure Disk Partitioning

Select Guided - Use Entire Disk and choose the 40 GB virtual disk. The resulting partition scheme, file system, and disk size are recorded in Section 6.



Step 8: Install OpenSSH Server

Select Install OpenSSH Server. No additional packages are installed unless instructed.





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Step 9: Complete Installation

Complete the installation, remove the installation media when prompted, and restart the server.

```
server [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
[ 25.961916] cloud-init[539]: Your identification has been saved in /etc/ssh/ssh_host_ecdsa_key
[ 25.962057] cloud-init[539]: Your public key has been saved in /etc/ssh/ssh_host_ecdsa_key.pub
[ 25.962190] cloud-init[539]: The key fingerprint is:
[ 25.962323] cloud-init[539]: SHA256:5444Qw3JocdW/EI8fH42fJLHfocsrJ3CSvR5JtvQ9 root@server01
[ 25.962459] cloud-init[539]: The key's randomart image is:
[ 25.962594] cloud-init[539]: +---[ECDSA 256]---+
[ 25.962730] cloud-init[539]: |o..+
[ 25.962866] cloud-init[539]: | +B
[ 25.963000] cloud-init[539]: |o 0..+ o +
[ 25.963141] cloud-init[539]: | = +. o s +
[ 25.963272] cloud-init[539]: | . + * . o s * o
[ 25.963401] cloud-init[539]: | . o + o . + .
[ 25.963537] cloud-init[539]: |ooo o
[ 25.963664] cloud-init[539]: |B .E. . o
[ 25.963801] cloud-init[539]: |o..+ o..+
[ 25.963937] cloud-init[539]: +---[SHA256]---+
[ 25.964032] cloud-init[539]: Generating public/private ed25519 key pair.
[ 25.964163] cloud-init[539]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 25.964293] cloud-init[539]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 25.964400] cloud-init[539]: The key fingerprint is:
[ 25.964522] cloud-init[539]: SHA256:8fHm2eqdW0IQ0ALVpJT7fChd01Adptch9L6EduE root@server01
[ 25.964642] cloud-init[539]: The key's randomart image is:
[ 25.964760] cloud-init[539]: +---[ED25519 256]---+
[ 25.964890] cloud-init[539]: |oE+ ..+
[ 25.964990] cloud-init[539]: |..+ +.o
[ 25.965141] cloud-init[539]: |* +B. . o
[ 25.965261] cloud-init[539]: | +o+ . o o
[ 25.965429] cloud-init[539]: |..+ +B S
[ 25.965568] cloud-init[539]: |..o.s
[ 25.965664] cloud-init[539]: |..+
[ 25.965781] cloud-init[539]: |..+
[ 25.965900] cloud-init[539]: |..+
[ 25.965921] cloud-init[539]: +---[SHA256]---+
ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:87:41:8b brd ff:ff:ff:ff:ff:ff
    altname enx08002787418b
    inet 192.168.2.12/24 metric 100 brd 192.168.2.255 scope global dynamic mngtaddr noprefroute
        valid_lft 86348sec preferred_lft 86348sec
    inet6 f107:525c:f037:c2a6::27ff:f077:418b/64 scope global dynamic mngtaddr noprefroute
        valid_lft 86398sec preferred_lft 14869sec
    inet6 f8a0::a90:27ff:f077:418b/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
geoff@server01:~$
```

6. Configuration Details

The table below summarizes the configuration applied during installation. Fields marked for completion should be filled in with the actual values obtained from the deployed virtual machine.

Item	Value
Hostname	server01
Network configuration	DHCP
Assigned IP address	[To be completed – insert actual IP from 'ip addr']



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Item	Value
SSH	Installed
Operating system	Ubuntu Server LTS

7. Verification Commands

Verify Hostname

```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
[ 25.961916] cloud-init[939]: Your identification has been saved in /etc/ssh/ssh_host_ecdsa_key
[ 25.962057] cloud-init[939]: Your public key has been saved in /etc/ssh/ssh_host_ecdsa_key.pub
[ 25.962190] cloud-init[939]: The key fingerprint is:
[ 25.962322] cloud-init[939]: SHA256:944rQM816z8yKEIAfM42fILhfocsrLJCSVR5JtVcB root@server01
[ 25.962459] cloud-init[939]: The key's randomart image is:
[ 25.962594] cloud-init[939]: +---[ECDSA 256]---+
[ 25.962733] cloud-init[939]: |o...
[ 25.962866] cloud-init[939]: |+.0
[ 25.963010] cloud-init[939]: |o 0+. o +
[ 25.963141] cloud-init[939]: | = * . o = *
[ 25.963272] cloud-init[939]: |. + *. oS*.o
[ 25.963401] cloud-init[939]: |.o.+ o. +.
[ 25.963557] cloud-init[939]: |ooo o . .
[ 25.963684] cloud-init[939]: |.B .E. . o
[ 25.963801] cloud-init[939]: |z.o . . .o.
[ 25.963917] cloud-init[939]: +---[SHA256]-----+
[ 25.964032] cloud-init[939]: Generating public/private ed25519 key pair.
[ 25.964163] cloud-init[939]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 25.964283] cloud-init[939]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 25.964400] cloud-init[939]: The key fingerprint is:
[ 25.964522] cloud-init[939]: SHA256:ER1m2xqkqk0I9G6VLVFPJ7IFChdD1ADptch9L6EQaUE root@server01
[ 25.964642] cloud-init[939]: The key's randomart image is:
[ 25.964768] cloud-init[939]: ++[ED25519 256]++
[ 25.964878] cloud-init[939]: |oEB* ..+.
[ 25.964990] cloud-init[939]: |.***+ .+o
[ 25.965141] cloud-init[939]: |* *B. . o
[ 25.965251] cloud-init[939]: |+=o+ . o o
[ 25.965429] cloud-init[939]: |.+o+oB S .
[ 25.965568] cloud-init[939]: |. ..oo.=
[ 25.965664] cloud-init[939]: | . +
[ 25.965781] cloud-init[939]: | .
[ 25.965836] cloud-init[939]: | .
[ 25.965921] cloud-init[939]: +---[SHA256]-----+
ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:07:41:0b brd ff:ff:ff:ff:ff:ff
    altname enx0002707410b
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86348sec preferred_lft 86348sec
    inet6 fd17:625c:f037:2::a0a:27ff:fe87:410b/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86350sec preferred_lft 14350sec
    inet6 fe80::a0a:27ff:fe87:410b/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
geoff@server01:~$
```

Expected result: server01



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Verify IP Address

ip addr

Test Internet Connectivity

ping -c 4 google.com

```
server [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
geoff@server01:~$ ping -c 4 google.com
PING google.com (142.250.205.238) 56(84) bytes of data:
64 bytes from maa05s28-in-f14.1e100.net (142.250.205.238): icmp_seq=1 ttl=255 time=10.7 ms
64 bytes from maa05s28-in-f14.1e100.net (142.250.205.238): icmp_seq=2 ttl=255 time=10.8 ms
64 bytes from maa05s28-in-f14.1e100.net (142.250.205.238): icmp_seq=3 ttl=255 time=10.3 ms
64 bytes from maa05s28-in-f14.1e100.net (142.250.205.238): icmp_seq=4 ttl=255 time=10.4 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 10.344/10.569/10.811/0.206 ms
geoff@server01:~$
```

Successful replies confirm internet connectivity.



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Update the Server

do apt update

sudo apt upgrade -y

```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help

apparmor      grub-pc-bin      linux-firmware-amd-graphics  linux-firmware-nvidia-graphics  python3-apport
apport        grub2-common     linux-firmware-amd-misc      linux-firmware-qlogic            python3-apt
apport-core-dump-handler  iproute2          linux-firmware-broadcom-wireless  linux-firmware-qualcomm-graphics  python3-distupgrade
base-files    keyboard-configur linux-firmware-intel-graphics  linux-firmware-qualcomm-misc      python3-netplan
cloud-init    libapparmor1      linux-firmware-intel-misc      linux-firmware-qualcomm-wireless  python3-problem-report
cloud-init-base  libaudit1-common  linux-firmware-intel-wireless  linux-firmware-realtek            rust-coreutils
console-setup  libaudit1         linux-firmware-marvell-prestera  mtdadm                            sg3-utils
console-setup-linux  libflashrom1      linux-firmware-marvell-wireless  mtd-news-config                   sg3-utils-udev
dmidecode     libfwupd3         linux-firmware-mediatek         netplan-generator                 snapd
fwupd         libnetplan1       linux-firmware-mellanox-spectrum  netplan.io                        sos
gir1.2-packagekit-glib-1.0  libpackagekit-glib2-1.0  linux-firmware-misc            packagekit                         thermald
grub-pc       libsgutils2-1.48  linux-firmware-netronome        python-apt-common                 ubuntu-release-upgrader-core

Not upgrading get due to phasing:
python3-software-properties  software-properties-common  ubuntu-kernel-accessories  ubuntu-minimal  ubuntu-server  ubuntu-server-minimal  ubuntu-standard

Summary:
Upgrading: 60, Installing: 0, Removing: 0, Not Upgrading: 7
Download size: 771 MB
Space needed: 2,729 kB / 14.2 GB available

Get:1 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 mtd-news-config all 14ubuntu6.2 [5,572 B]
Get:2 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 base-files amd64 14ubuntu6.2 [74.9 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 console-setup-linux all 1.237ubuntu3.1 [6,206 kB]
Get:4 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 console-setup all 1.237ubuntu3.1 [103 kB]
Get:5 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 keyboard-configuration all 1.237ubuntu3.1 [211 kB]
Get:6 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libaudit1-common all 1:4.1.2-1ubuntu0.1 [4,958 B]
Get:7 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libaudit1 amd64 1:4.1.2-1ubuntu0.1 [55.1 kB]
Get:8 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 rust-coreutils amd64 0.10.0-1ubuntu2~26.04.1 [4,515 kB]
Get:9 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 iproute2 amd64 6.19.0-1ubuntu1.1 [1,198 kB]
Get:10 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libapparmor1 amd64 5.0.2-0ubuntu1~26.04.1 [51.3 kB]
Get:11 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-netplan amd64 1.2-1ubuntu5.1 [24.8 kB]
Get:12 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 netplan-generator amd64 1.2-1ubuntu5.1 [65.1 kB]
Get:13 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 netplan.io amd64 1.2-1ubuntu5.1 [69.6 kB]
Get:14 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libnetplan1 amd64 1.2-1ubuntu5.1 [132 kB]
Get:15 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python-apt-common all 3.1.0ubuntu1.1 [17.0 kB]
Get:16 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-apt amd64 3.1.0ubuntu1.1 [165 kB]
Get:17 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apparmor amd64 5.0.2-0ubuntu1~26.04.1 [380 kB]
Get:18 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 dmidecode amd64 3.6-2ubuntu1 [72.8 kB]
Get:19 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-release-upgrader-core all 1:26.04.25 [24.9 kB]
Get:20 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-distupgrade all 1:26.04.25 [106 kB]
Get:21 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-problem-report all 2.34.1-0ubuntu0.1 [17.4 kB]
Get:22 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-apport all 2.34.1-0ubuntu0.1 [36.4 kB]
Get:23 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apport-core-dump-handler all 2.34.1-0ubuntu0.1 [9,674 B]
Get:24 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apport all 2.34.1-0ubuntu0.1 [83.9 kB]
Get:25 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 cloud-init-base all 26.1-0ubuntu3~26.04.1 [587 kB]
Get:26 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 cloud-init all 26.1-0ubuntu3~26.04.1 [2,110 B]
Get:27 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libflashrom1 amd64 1.6.0-2ubuntu1.1 [226 kB]
Get:28 http://ph.archive.ubuntu.com/ubuntu resolute-updates/main amd64 fwupd amd64 2.1.1-1ubuntu3.1 [12.7 MB]
11% [28 fwupd 4,108 kB/12.7 MB 32%]

2,222 kB/s 5min 38s
Right Ctrl
```



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Verify SSH

systemctl status ssh

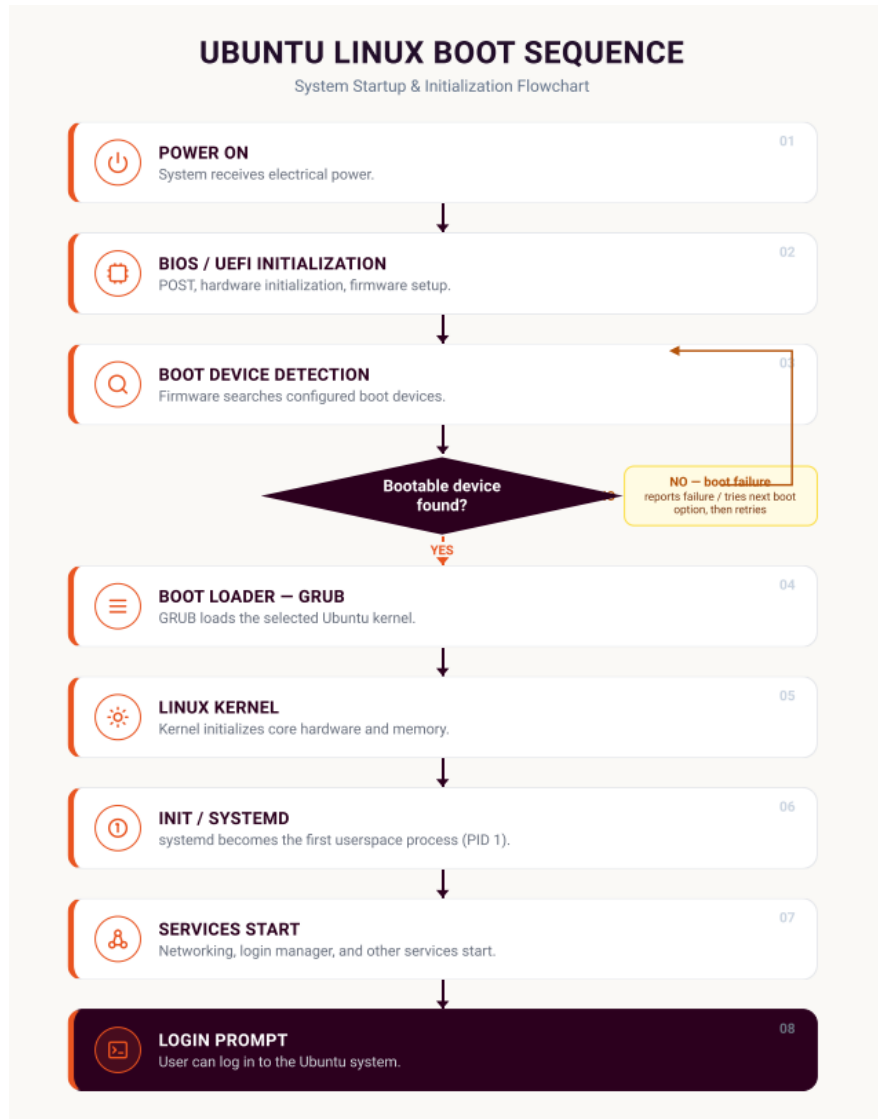
Expected result: active (running)

```
server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
geoff@server01:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Fri 2026-09-25 01:17:08 UTC; 5min ago
   Invocation: ba50e96d6594c08abd291e1ec54015
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
         man:sshd_config(5)
   Process: 1238 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1241 (sshd)
   Tasks: 1 (Limit: 1083)
   Memory: 1.6M (peak: 2.2M)
   CPU: 38ms
   CGroup: /system.slice/ssh.service
           └─1241 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 25 01:17:07 server01 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 25 01:17:08 server01 sshd[1241]: Server listening on 0.0.0.0 port 22.
Sep 25 01:17:08 server01 sshd[1241]: Server listening on :: port 22.
Sep 25 01:17:08 server01 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
geoff@server01:~$
```



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8. BIOS vs UEFI Comparison

A full comparison of BIOS and UEFI firmware, including a detailed comparison table and an explanation of why UEFI has largely replaced BIOS in modern systems, is provided in the accompanying document BIOS_vs_UEFI.pdf.



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9. Ubuntu Boot Process

The Ubuntu Server boot process consists of eight stages, from power-on through to the login prompt:

- 1 Power On
- 2 BIOS/UEFI Initialization
- 3 Boot Device Detection
- 4 Boot Loader (GRUB)
- 5 Linux Kernel
- 6 init / systemd
- 7 Services Start
- 8 Login Prompt

A visual flowchart of this process is provided in the accompanying document [BootProcessFlowchart.pdf](#).

10. Windows Server Installation Summary

Windows Server Evaluation was installed in a separate virtual machine as a bring-home activity. The required tasks were:

- Install the operating system.
- Assign a computer name.
- Create an administrator password.
- Log in successfully.

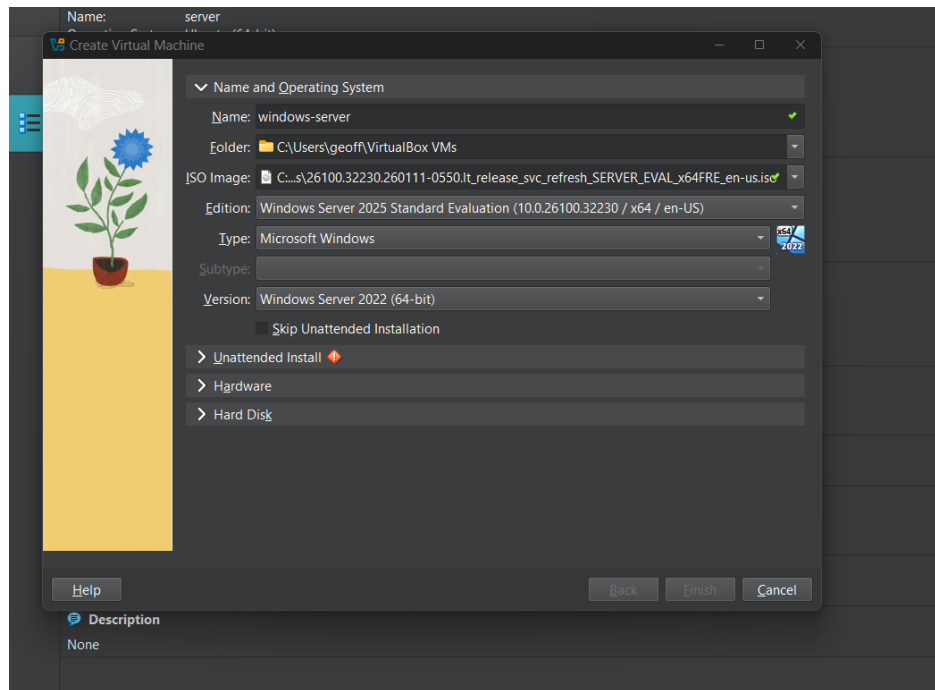


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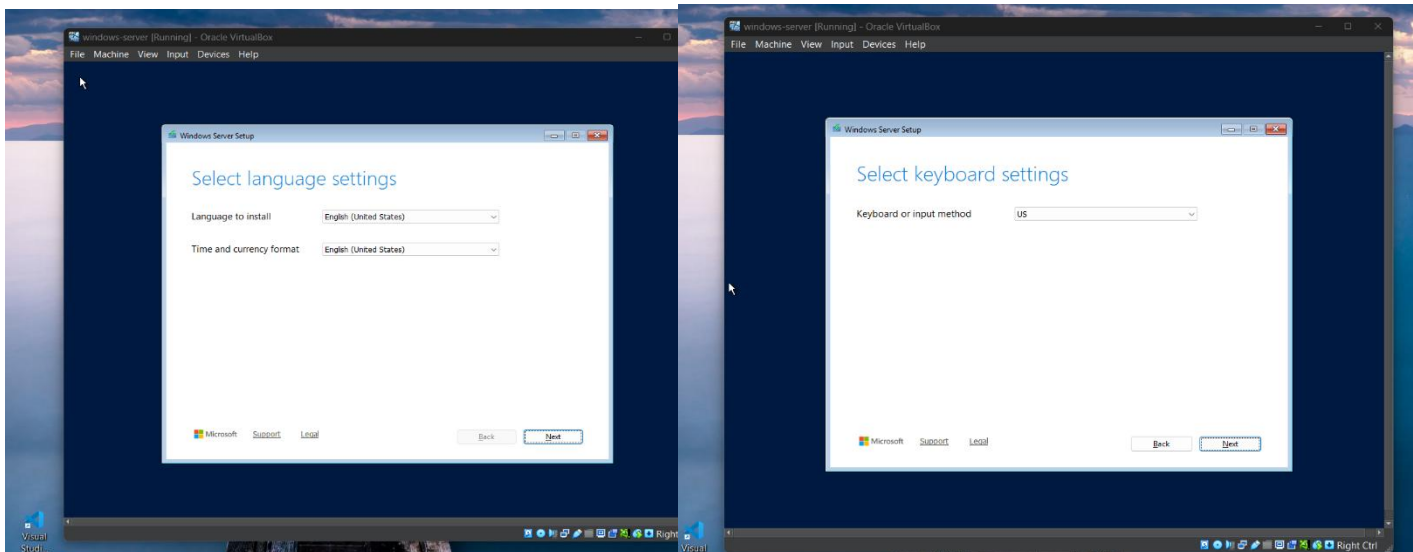


- Capture screenshots of the installation.

Step 1 - Create the Virtual Machine



Step 2 - Select Language and Keyboard

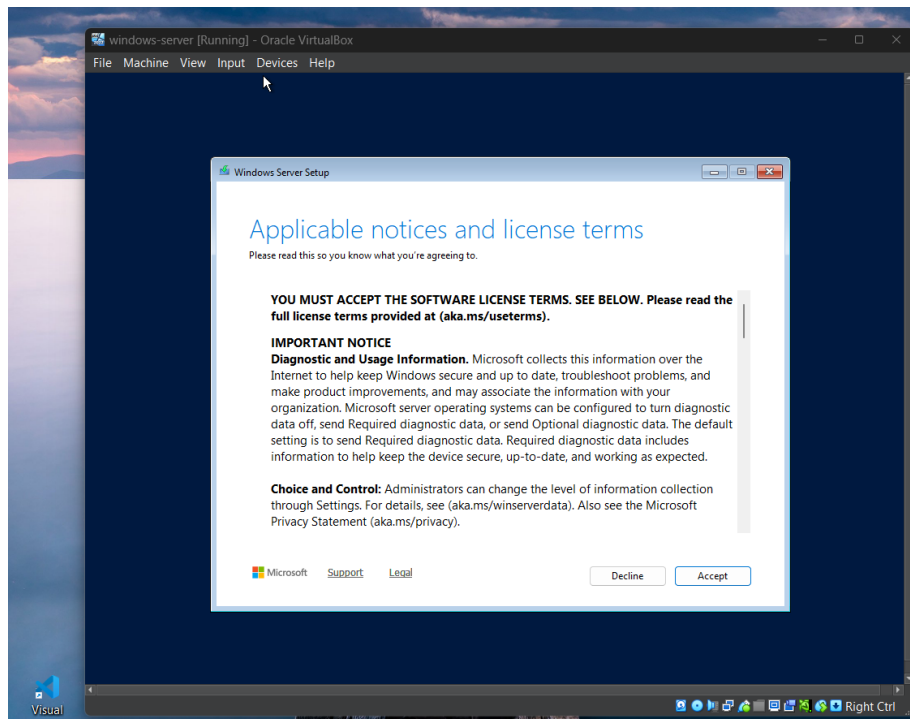




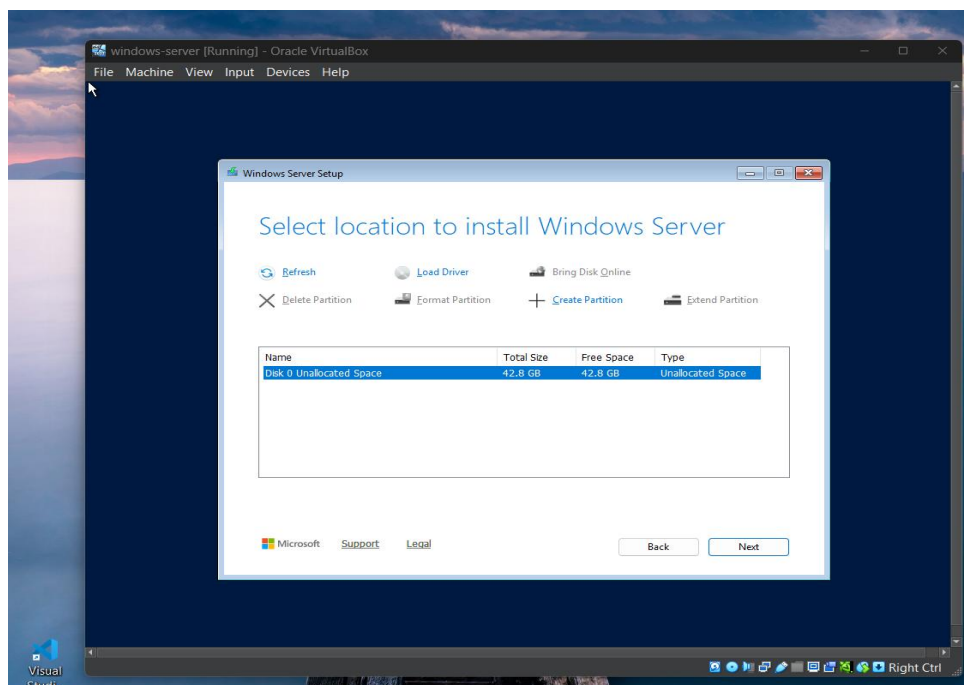
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Step 3 - Accept the License Terms



Step 4 - Select the Virtual Disk

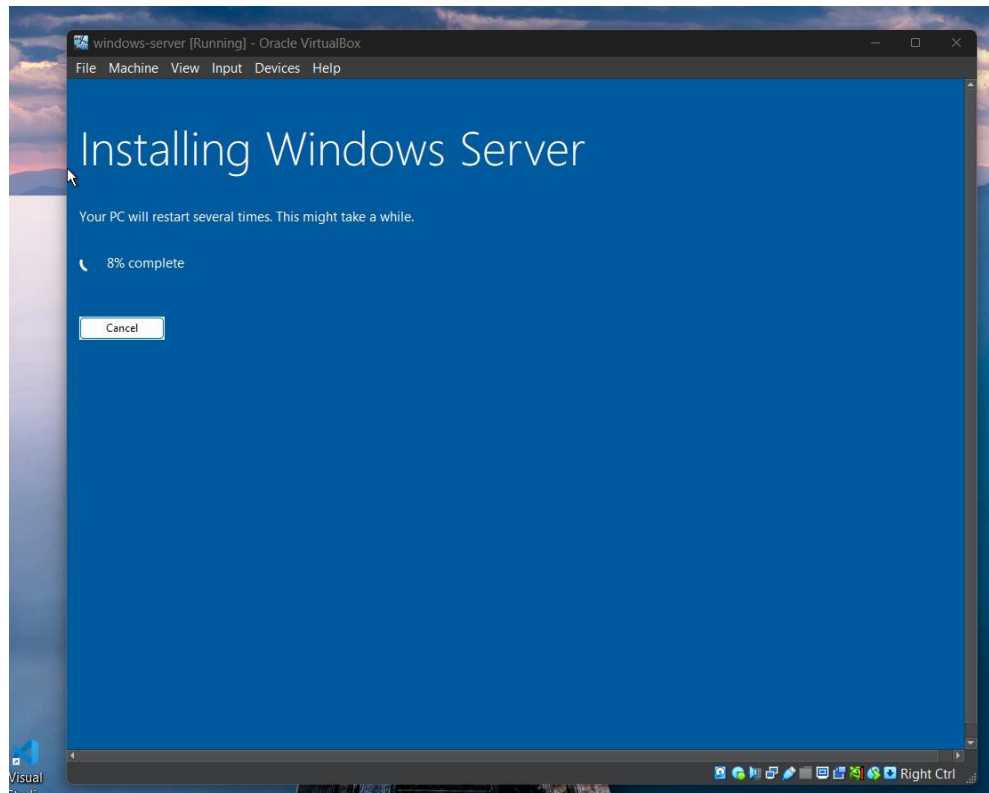




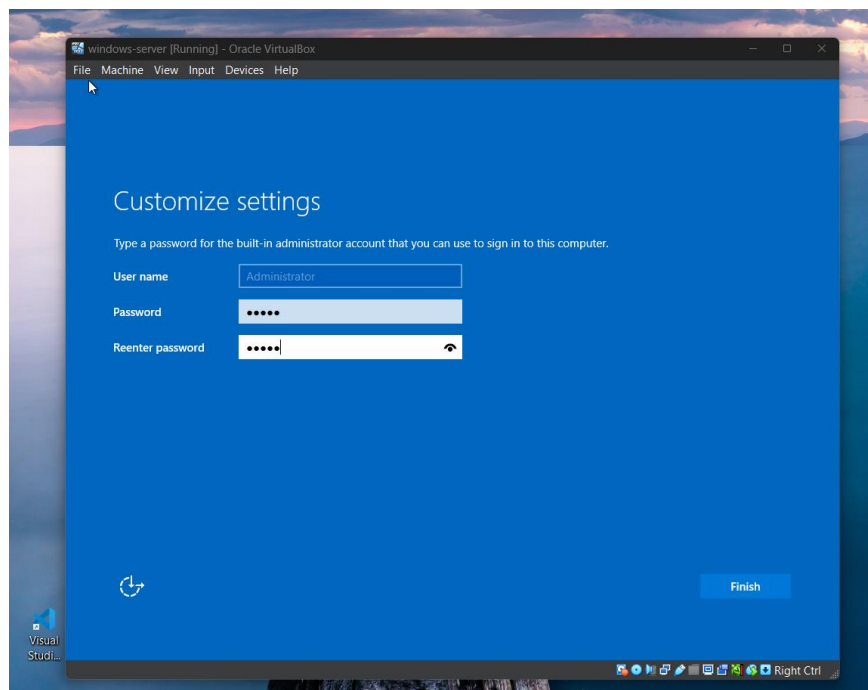
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Step 5 - Wait for Installation



Step 6 - Set the Administrator Password





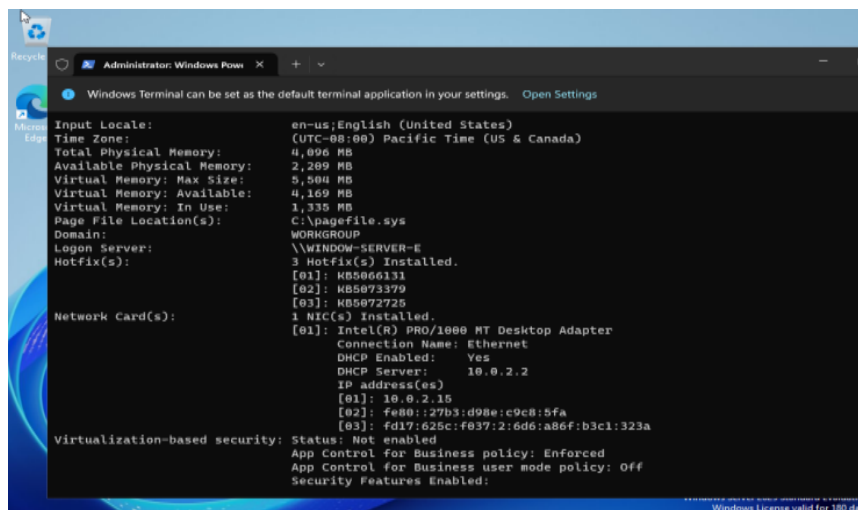
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Step 7 - Log In



Step 8 – Details





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11. Operating System Comparison

Windows Server, Ubuntu Server, and Rocky Linux were compared across licensing, user interface, package management, security, performance, and typical enterprise use cases.

Category	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Commercial, paid per-license/CAL model	Free and open source	Free and open source
User Interface	GUI available (Desktop Experience) plus CLI	CLI-first, minimal GUI options	CLI-first, minimal GUI options
Package Management	MSI / Windows Update / PowerShell	APT (.deb packages)	DNF/YUM (.rpm packages)
Security	Built-in Defender, Group Policy, regular patch cycle	AppArmor, frequent security updates, strong community patching	SELinux, enterprise-grade RHEL-compatible security model
Performance	Higher resource overhead, GUI-	Lightweight, efficient for server workloads	Lightweight, efficient, optimized for stability



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Category	Windows Server	Ubuntu Server	Rocky Linux
	dependent workloads		
Typical Enterprise Use	Active Directory, .NET apps, Windows-native services	Web hosting, containers, cloud-native workloads	RHEL-compatible production servers, long-term stability needs

Advantages and Disadvantages

Windows Server – Advantage: tight integration with Windows-based enterprise tools and Active Directory. Disadvantage: licensing cost and higher resource usage.

Ubuntu Server – Advantage: free, large community, frequent updates, well suited to cloud and container workloads. Disadvantage: shorter default support window compared to some enterprise-focused distributions.

Rocky Linux – Advantage: binary compatibility with RHEL and long-term stability, popular for production servers. Disadvantage: smaller desktop/GUI ecosystem and slower adoption of bleeding-edge packages.



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12. Problems Encountered

During the installation process, I encountered several challenges. The first problem was a misunderstanding between Ubuntu and Ubuntu Server. At first, I thought that the Ubuntu operating system I had installed was already the same as the server version. Because of this misunderstanding, I decided to uninstall the existing Ubuntu operating system and replace it with Ubuntu Server. This required me to download the Ubuntu Server installation file again and perform a new installation, which took additional time. Another problem occurred when I forgot the password that I had configured for the server. Since I did not know how to properly change or reset the password, I decided to delete the existing server and reinstall Ubuntu Server from the beginning.

The last problem was related to the internet connection. During the installation and downloading process, the internet connection experienced fluctuations, which caused some downloads and installation processes to fail or become interrupted. Since some installation files were large, an unstable connection meant that I sometimes had to repeat the download or installation process. These problems made the installation take longer than expected and showed me the importance of proper preparation before starting a server installation.

13. Solutions Implemented

To address the problems I encountered, I first learned to properly distinguish between Ubuntu Desktop and Ubuntu Server and to verify the required operating system before performing an installation. I also learned that when a password is forgotten, reinstalling the entire server is not always necessary because there are procedures available for resetting or changing the



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password. Understanding basic user and password management can help avoid unnecessary deletion of the server and prevent the loss of existing configurations or files.

For the internet connection problem, I made sure to check the network connection before starting the installation and downloading the required files. I also allowed important files to finish downloading completely before proceeding with the installation. When the connection was unstable, I waited for it to become more reliable before continuing. These solutions helped reduce installation errors and unnecessary repetition of the process.

14. Lessons Learned

Through this activity, I learned the importance of understanding the operating system and installation requirements before making major changes to a server. My initial misunderstanding between Ubuntu and Ubuntu Server resulted in an unnecessary reinstallation, which taught me to carefully read the instructions and verify the correct operating system before beginning. I also learned that forgetting a password does not necessarily require deleting and reinstalling the entire server. Learning proper password and user management procedures can make troubleshooting more efficient and prevent unnecessary loss of configurations or files.

I also learned that a stable internet connection is important when installing a server because the process may require downloading large installation files and additional packages. An unstable connection can interrupt the installation and cause errors that require the process to be repeated. So all of this experience improved my troubleshooting and basic system administration skills. It taught me to identify the actual cause of a problem first, research or



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apply the appropriate solution, and avoid immediately reinstalling the system whenever an issue occurs.

15. References

Ubuntu Documentation Team. Ubuntu Server Guide. Canonical Ltd.

<https://ubuntu.com/server/docs>